

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-07

Autonomous Multi-Agent Drone Swarm Protocol for Disaster Response

Domain: Robotics / Swarm Intelligence / Autonomous Systems

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Formulate a decentralized peer-to-peer swarm mesh communication and navigation protocol enabling autonomous search-and-rescue mapping in GPS-denied catastrophe zones.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] Decentralized flocking and collision avoidance based on Reynolds swarm heuristics and APF.

[REQ-2] Visual SLAM (Simultaneous Localization and Mapping) for indoor and collapsed structure navigation.

[REQ-3] Real-time thermal infrared survivor detection and victim geolocation tagging.

[REQ-4] Self-healing dynamic ad-hoc mesh network maintaining relay connectivity with base camp.

[REQ-5] Automated payload drop trajectory calculation for critical emergency medical kits.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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